

Khulna University of Engineering & Technology  
Department of Urban and Regional Planning  
BURP 2<sup>nd</sup> Year 2<sup>nd</sup> Term Examination, 2016.  
**CE 2241**  
(Principles of Environmental Engineering)

Full Marks: 210

Time: 3 hrs

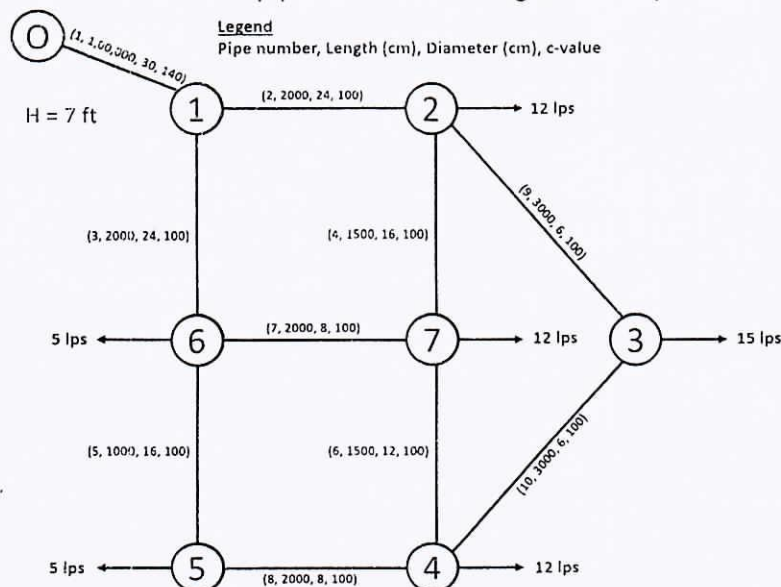
- N.B.** i) Figures in the right margin indicate full marks.  
ii) Answer any **three** questions from each section in separate script.  
iii) Assume reasonable values for any missing data.

**Section – A**

1. a) Write down the basic assumptions about the hydraulic conditions in the aquifer. (10)  
b) When is backwashing needed? Write down the remedial measures to prevent cracking of filters and formation of mudballs. (05)  
c) As the Chief Operator on behalf of the Mayor and City Council estimate the projected annual operating budget (rounded to the nearest \$100) for chlorination at a booster pump station. (15)  
Data: The booster pumps run on a continuous 12 hour alternating cycle (both pumps identical). The pump is 125 HP, with an efficiency of 82%, pumping into a 18 inch D.I. pipe. The elevation of the pumps is 4118 feet and the reservoir they are pumping to is 4356 feet (floor). The tank is 155 feet in dia with an overflow at 25 feet above the floor. The chlorine demand is 1.3 mg/L with a required free chlorine residual 0.6 mg/L and the cost of chlorine is 43 cents per pound.  
d) Describe “ILLER BANKASI METHOD” for estimating future populations. (05)
2. a) What are the objectives of water supply? Mention the elements of water supply system. (06)  
b) What are the factors influencing per capita water consumption? Write down the water quality criteria for drinking purpose. (10)  
c) State the effects of impurities present in water. (08)  
d) Estimate from following statistics the population expected in 2025. Use least square parabola method. (11)

| Year                   | 1970 | 1980 | 1990 | 2000 | 2010 |
|------------------------|------|------|------|------|------|
| Population (thousands) | 30   | 38   | 48   | 59   | 70   |

3. Calculate the flow in each of the pipes of the following network (at least two trials) (35)



4. a) A water supply system company delivered an annual quantity of 10,000,000 m<sup>3</sup> to a city of 1.2 million inhabitants assuming an average leakage of 20%. On the maximum consumption day, the registered delivery was as follows: (25)

|                |      |      |      |     |      |      |      |      |      |      |      |      |
|----------------|------|------|------|-----|------|------|------|------|------|------|------|------|
| Hour           | 1    | 2    | 3    | 4   | 5    | 6    | 7    | 8    | 9    | 10   | 11   | 12   |
| m <sup>3</sup> | 989  | 945  | 902  | 727 | 844  | 1164 | 1571 | 1600 | 1775 | 1964 | 2066 | 2110 |
| Hour           | 13   | 14   | 15   | 16  | 17   | 18   | 19   | 20   | 21   | 22   | 23   | 24   |
| m <sup>3</sup> | 1600 | 1309 | 1091 | 945 | 1062 | 1455 | 1745 | 2139 | 2110 | 2037 | 1746 | 1018 |

Determine: a) diurnal peak factors for the area. b) The maximum seasonal variation factor. c) Diurnal consumption factors. d) Specific water demand in the distribution area including leakage. e) Calculate the demand in 20 years time if the assumed annual demand growth is 2.5%. Compare the results by applying the linear and exponential models.

- b) Define potable water and well hydraulics. Enumerate the general considerations for planning and design of a water supply system. (10)

### Section – B

5. a) What is the global sanitation scenario with respect to population coverage? What is the sanitation in Bangladesh? (10)
- b) Define sanitation. What are the objectives of sanitation? (06)
- c) Draw a schematic diagram of a compost latrine showing different components. (07)
- d) Design a VIP latrine for a family of ten. The family uses water for anal cleansing. The GWT is only 2m below the ground surface. (12)
6. a) Draw a schematic diagram showing various components of a septic tank. Write down the process take place within the septic tank. Why the inlet of a septic tank is usually made with a tee junction? (12)
- b) Distinguish communal sanitation and public toilet facilities. What is the primary reason for failures of such communal and public sanitation facilities? How can these services be made sustainable? (10)
- c) A common septic tank is to be designed for three adjoining houses with a total of 21 users. The average wastewater flow rate is 180 lpcd. Design a suitable septic tank. (13)
7. a) What are the basic functional elements of a conventional sewerage system? Write down the types of sewage collection system. (06)
- b) Describe the basic principles of human waste treatment process. (12)
- c) Discuss the algae-bacteria symbiosis in a facultative pond. (07)
- d) Define various types of waste stabilization ponds. Describe the primary functions of each type giving suitable diagrams. (10)
8. a) What factors are considered in the physical analysis of solid waste composition? Compare the composition of solid wastes generated in Bangladesh and Europe and discuss reasons for the variation in composition. (10)
- b) Describe the basic process of aerobic composting with a suitable flow diagram. Briefly discuss the important parameters that need careful consideration for efficient aerobic composting. Comment on the applicability of composting in Bangladesh. (15)
- c) What are the various collection systems of solid wastes? Which type is the most appropriate for the city of Khulna and why? What factors should be considered in designing the optimum solid waste collection route? (10)



Khulna University of Engineering & Technology  
Department of Urban and Regional Planning  
BURP 2<sup>nd</sup> Year 2<sup>nd</sup> Term Regular Examination, 2016  
**Hum 2217**  
(Public and Local Government Finance)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.  
ii) Figures in the right margin indicate full marks.

**Section – A**

1. (a) What is public finance? Distinguish between public finance and private finance. (10)  
(b) Describe the importance of studying Public Finance in the Department of Urban and Regional Planning. (15)  
(c) Explain the functions of public finance in a country. (10)
2. (a) Differentiate tax revenue from non-tax revenue. (10)  
(b) Define tax. Make a list of advantages and disadvantages of direct and indirect taxes. (10)  
(c) Explain the principles of a good tax system. (15)
3. (a) What is public debt? Explain various forms of public debt. (15)  
(b) Describe the burdens of internal debt with necessary example. (10)  
(c) Explain proportional tax and regressive tax with necessary example. (10)
4. Write short notes on the followings (any four): (35)
  - i) Private, public and free goods
  - ii) Sales tax and income tax
  - iii) Value added tax and tariff
  - iv) Bangladesh Municipal Development Fund
  - v) Property tax and corporate tax



**Section – B**

5. (a) Define budget. Describe various types of public budget. (15)  
(b) Discuss the backdrop of public budget preparation. (10)  
(c) What are the main sources of revenue of local government in Bangladesh? Discuss in brief. (10)
6. (a) Illustrate the concept 'common resource' with necessary example. Without the intervention of government, whether the people will use this good too much or too little? Justify your position. (15)  
(b) What is decentralization? Discuss the importance of decentralization in development. (10)  
(c) Explain the functions of a city corporation. (10)
7. (a) What was the first initiative as regards the reform in local government of Bangladesh in the post-independence period? (10)  
(b) What is local government? Explain the structure of local government of Bangladesh. (10)  
(c) State the structure/composition and function of a *Union Parishad*. (15)
8. (a) Explain the role of deficit financing in underdeveloped country like Bangladesh. (20)  
(b) Discuss the effects of deficit financing in a country. (15)

Khulna University of Engineering & Technology  
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BURP 2<sup>nd</sup> Year 2<sup>nd</sup> Term Regular Examination, 2016  
**URP 2251**  
(Landscape Planning and Design)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.  
ii) Figures in the right margin indicate full marks.

**Section – A**

1. (a) Define landscape. (05)  
(b) Discuss with example the needs of studying landscape planning and design for the students of URP. (15)  
(c) Distinguish landscape feature, landscape element and landscape principle with example and necessary drawings. (15)
2. (a) What is texture? (05)  
(b) Discuss different types of texture with example and drawing. (10)  
(c) Explain the color schemes used in landscape planning and design. (10)  
(d) Discuss different forms of unity as landscape principle. (10)
3. (a) Evaluate the nine landscape principles that are generally followed in landscape plan and design of a city. (11)  
(b) Discuss the landscape planning and design in terms of general challenges, goal, plant characteristics and design solutions with necessary sketch on any two of the followings: (24)  
i) Front entry ii) Along walls iii) Along sidewalks.
4. (a) Define ecology in terms of landscape. (05)  
(b) Discuss the Carbon Cycle with relationship of different elements in necessary drawings. (10)  
(c) Write short notes on the followings (any four): (20)  
i) Different lines and their properties  
ii) Proportion in landscape planning and design  
iii) Tree forms in landscape  
iv) Trees in controlling noise pollution  
v) Trees in controlling solar radiation and temperature.

**Section – B**

5. (a) Describe different theory of landscape for specialized areas with necessary examples and sketches. (20)  
(b) What do you mean by urban open and green space? Describe different types of urban open and green spaces. (15)
6. (a) Why is water considered as a significant resource in landscape planning? (05)  
(b) How can we get the maximum benefit from water features in a landscape? (20)  
(c) How is a residential landscape planning different from other forms of landscape planning? Explain. (10)
7. (a) Why is scale an important factor in landscape planning and design? (05)  
(b) Discuss different steps to develop a residential landscape with sketches. (15)  
(c) Explain different types of conservation principles in landscape. (15)
8. (a) Draw an explanatory connection between residential, commercial and recreational environments. (15)  
(b) Write short notes on the followings (any four): (20)  
i) Streetscape ii) Visible landscape iii) Buffer zone iv) Urban Fabric v) Urban scale.

Khulna University of Engineering & Technology  
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**URP-2281**  
(GIS and Remote Sensing)

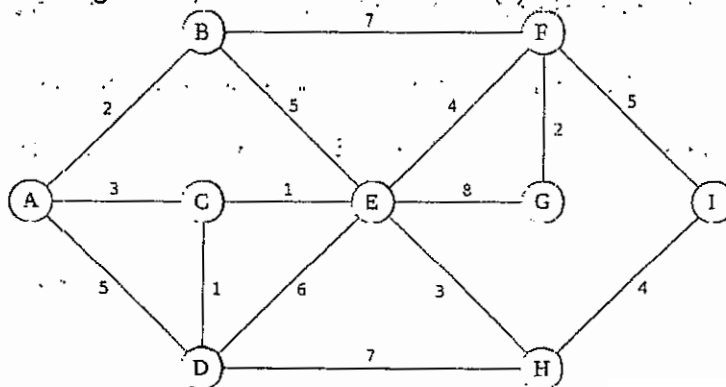
Full Marks: 210

Time: 3.00 hrs

- N.B. i) Answer any THREE questions from each section in separate scripts.  
ii) Figures in the right margin indicate full marks.

**Section – A**

1. (a) "Knowing where things are, and why, is essential to rational decision making". (10)  
Explain the statement with examples.  
(b) Show the relationship between Spatial Data Modeling and Database (05)  
Management System.  
(c) "Spatial is special". Explain your reasons as to why you agree or disagree with (05)  
this statement.  
(d) Provided are the surge values of a river and DEM of a nearby area to the river, (05)  
can one find the places where flooding might occur?  
(e) What are the challenges associated with the usage of GIS? (10)
2. (a) A wildlife photographer wants to capture some shots. What data/layer should (10)  
he be concerned with so as to find the location of exotic animals? Explain.  
(b) "One of the components of GIS is people, not only that but everybody is a (05)  
provider". Explain.  
(c) Show the differences between DBMS and RDBMS with the help of an (05)  
example.  
(d) Explain the need for GIS in an academic institution like KUET. (05)  
(e) Provide the names of some Major GIS User Organizations in Bangladesh. (10)
3. (a) Provide information about the following types of maps. (15)  
(i) Soil Reaction  
(ii) Crop Suitability  
(iii) Flood Prone Area  
(b) There is a strong relationship between GIS and Digital Bangladesh. Explain. (10)  
(c) What are the major drawbacks of Spatial Database Infrastructure? (05)  
(d) Desktop, Server and Web, why does GIS require such an Integrated Systems (05)  
Approach?
4. (a) Explain the components of Semivariogram with the help of a diagram. (05)  
(b) Using Dijkstra's Algorithm, find the shortest route(s) from A to I. (15)



- (c) Explain "Thiessen Polygon" and its construction. (10)
- (d) Explain the "Travelling salesperson problem". (05)

## Section – B

5. (a) Write down your understanding about the need for Remote Sensing and Photogrammetric education. (05)
- (b) Classify the different types of satellites. Differentiate between Geostationary and Geosynchronous satellites. (10)
- (c) Write down the name of the corrections for satellite image processing (it may be pre-processing or post-processing) with proper explanation based on the following different types of error. (10)
- i. Terrain effect due to undulation on the earth surface
  - ii. Absence of sufficient local solar illumination
  - iii. Satellites or aircraft not staying at a constant altitude or by sensors deviating from the primary focus plane
  - iv. Variation of the surface reflectance on the same surface due to bad weather
- (d) Briefly explain the remote sensing sensors and their different types. (10)

6. (a) Compare between two sets of satellite imageries having different types of different resolutions in the following. Explain with proper example. (10)

| Satellite Image (A)                                                                                                                                                                             | Satellite Image (B)                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• High spatial resolution (0.5 - 4 m)</li> <li>• High spectral resolution (1- 3 bands)</li> <li>• Medium Radiometric resolution (8 bit image)</li> </ul> | <ul style="list-style-type: none"> <li>• High spatial resolution (0.5 - 4 m)</li> <li>• Medium spectral resolution (4 - 15 bands)</li> <li>• High Radiometric resolution (16 bit and above)</li> </ul> |

- (b) Explain remote sensing satellite with proper example. (07)
- (c) Briefly explain the tasks included in image processing of remote sensing. (08)
- (d) Explain the relationships between Instantaneous Field of View (IFOV) and Angular Field of View (AFOV) with providing proper figure. How does swath width of the satellite images depend upon the Angular Field of View (AFOV)? (10)
7. (a) What is your understanding about Photogrammetry? Differentiate between Photogrammetry and Remote Sensing technology. (10)
- (b) Assume that the distance between two points was measured to be 83.33 mm on a vertical photograph and 125 mm on a map. If the surveying ground distance between the same two points is 3000 m, what will be the photo scale and map scale and the scale reciprocals of the photograph and the map? (10)
- (c) When do you use True Color Combination (TCC) and False Color Combination (FCC) in remote sensing analysis and why? (10)
- (d) Why aerial photographs are called the first form of remote sensing imagery? (05)
8. (a) "Photography is the first part of photogrammetry where metrology is the second part" – Briefly explain with proper example. (05)
- (b) Write down the advantages and disadvantages of Photogrammetry. (08)
- (c) Explain any three applications of the Photogrammetry of the following: (12)
- i. Topographic Mapping
  - ii. Detailed Survey of Historic Buildings
  - iii. Terrestrial, Industrial and Scientific Uses
  - iv. Cadastral Plans
- (d) Explain the relationship between photo scale and map scale with proper figure. (10)



Khulna University of Engineering & Technology  
Department of Urban and Regional Planning  
BURP 2<sup>nd</sup> Year 2<sup>nd</sup> Term Examination, 2016  
**URP 2291**  
(Statistics for Planners-II)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any three questions from each section in separate script.  
ii) Figures in the right margin indicate full marks.  
iii) Necessary charts and graphs will be supplied.  
iv) Assume reasonable value for any missing data

**Section – A**

1. (a) Define correlation and regression. Write down three properties for each of them. (08)  
(b) Using the following data find the coefficient of linear correlation by using (20)  
(i) Product-moment and (ii) Rank correlation method.

|        |   |   |   |    |    |
|--------|---|---|---|----|----|
| Day    | 2 | 4 | 6 | 8  | 20 |
| Weight | 3 | 6 | 8 | 10 | 4  |

Which result is acceptable, why?

- (c) Show that the correlation coefficient is independent of scale of measurement [forbidden to consider transformation of origin]. (07)
2. (a) What is test of hypothesis? (05)  
(b) Define the followings with example: (i) Null hypothesis and Alternative hypothesis. (ii) Type I error and Type II error. (12)  
(c) What do you mean by level of confidence and level of significance? (12)  
(d) Write down the use of t-test. (06)
3. (a) Discuss briefly about the importance of statistics for planners. (08)  
(b) A radio shop sells on average 200 radios per day, with a standard deviation of 50 radios. After an expensive advertising campaign, the management will compute the average sales for the next 258 days to see whether an improvement has occurred. Assume that the daily sales of radios is normally distributed.  
(i) Write down the null and alternative hypothesis.  
(ii) Test hypothesis at 5% level of significance if  $\bar{x} = 126$ .  
(iii) How large must  $\bar{x}$  be in order that the null hypothesis is rejected at 5% level of significance.  
(c) Show that the Normal distribution is the particular case of  $\chi^2$  - (Chi-square) distribution when  $n = 1$ . (11)

4. (a) What is  $\chi^2$  - (Chi-square) test? The following table shows the number of recruits taking a preliminary and a final test in car driving. (20)

|       |      | Preliminary |      |
|-------|------|-------------|------|
|       |      | Pass        | Fail |
| Final | Pass | 605         | 135  |
|       | Fail | 195         | 65   |

Use  $\chi^2$  -test to discuss whether there is any association between the results of the preliminary and those of final test.

- (b) Explain the probability density function (p.d.f) of  $\chi^2$  - (Chi-square) distribution with  $n$  degrees of freedom. (15)

## Section – B

5. You have the following information from WFD. (35)

| Year                | 2001 | 2002 | 2003 | 2004 | 2005 | 2008 |
|---------------------|------|------|------|------|------|------|
| Rainfall (cm)       | 7    | 10   | 7    | 9    | 12   | 5    |
| Cost of damage (tk) | 5    | 8    | 6    | 7    | 10   | 3    |

- Find a simple Linear Regression model of the cost of damages on Rainfall.
- Find a quadratic (Polynomial) Regression model of the cost of damages on Rainfall.
- If the standard error is less than 10 and the coefficient of determination is greater than 0.9, then the linear model is applicable for forecasting. Is the above linear regression model good enough for forecasting, why?
- Estimate the cost of damage when rainfall is 20 cm by using relatively better model of the two considered above.

6. (a) Derive simple linear regression model. (09)

- (b) Plot the following numerical data on a graph paper. Hence establish an appropriate curvilinear mathematical regression model of Sft on income. (17)

| Place  | BD | IND | UK | CN | AM | IT |
|--------|----|-----|----|----|----|----|
| Income | 2  | 3   | 5  | 4  | 7  | 6  |
| Sft    | 2  | 4   | 13 | 8  | 25 | 19 |

Hence estimate Sft for the given income of IND according to the model you considered.

- (c) Show that the variance of estimated  $Y_e$  is  $r^2$  times the variance of observed  $y$  [i.e.  $s_{Y_e}^2 = r^2 s_y^2$ ]. (09)

7. (a) Show that the regression's coefficient is independent of the origin of measurement [forbidden to consider scale factor]. (07)

- (b) Represent the summary of well-known population growth models; also mention their graphical views. (13)

- (c) If you have the following information of a city. (15)

| Year       | 2005     | 2015     |
|------------|----------|----------|
| Population | 1,20,000 | 1,35,000 |

- Find the arithmetic growth rate of the city.
- Forecast the population at 2018.
- Derive the general equation regarding arithmetic growth rate.
- Find out the time when the population will be double of the population at 2015.

8. (a) After formulation derive the Logistic curve population forecasting model. (10)

- (b) The population of a city in consecutive years i.e. 2000, 2005 and 2010 are 100,000; 200,000; and 350,000 respectively. Determine (11)

- Saturated population.
- The equation of logistic curve.
- The expected population in 2025.

- (c) Predict the population for the year 2020, 2030 and 2050 from the following population by using incremental increasing method. (08)

| Year       | 1950  | 1960   | 1970   | 1980   | 1990   | 2000   | 2010  |
|------------|-------|--------|--------|--------|--------|--------|-------|
| Population | 85800 | 101500 | 120150 | 169200 | 200300 | 220000 | 26000 |

- (d) What do you mean by (i) Cohort study and (ii) Household panel survey? (06)